Hydrological response of a coastal dune aquifer to plantation clearfell and dune scraping: a BACI analysis of a 21-year manual record

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Abstract

Study region: Newborough Warren, Anglesey, north-west Wales — a single coastal sand body carrying dune-slack habitat of European conservation importance, a mature Corsican pine plantation on the stabilized back dunes, and a retreating Caernarfon Bay shoreline, monitored by a 21-year manual dipwell record (88 wells, 2005–2026).

Study focus: Two restoration interventions are evaluated in a Before–After–Control–Impact design: a 2015 dune scraping at one slack, and an experimental ~8 ha clearfell in December 2017. Both are analysed by ANCOVA with a cumulative-water-balance covariate, a distance-weighted scraping term and an coastal-drift control, against five control tiers including in-situ unfelled forest. Effects are reported for the mean water table and summer minimum depth, the metric governing slack viability, then placed on a common site-integrated footing alongside the site's background drivers.

New hydrological insights for the region: The clearfell raised the mean water table at the felled well by +0.113 m (95% CI [+0.042, +0.183], $p = 0.002$) but changed summer minima at no tier: canopy removal is a winter-recharge intervention and does not relieve the summer drought that constrains slack viability. Scraping delivered a larger, directly ecological gain: +0.195 m in summer minimum depth ($p = 0.004$), but it is mechanistically identical to coastal erosion, draining its surroundings and carrying a chase-inland cascade risk that makes siting decisive. Both interventions are second-order against coastal-retreat drawdown.

Highlights

- The 2017 clearfell raised the mean water table +0.113 m; summer minima unchanged.

- Dune scraping lifted the summer minimum +0.195 m — the larger ecological gain.
- Scraping drains its surroundings like coastal erosion: a chase-inland cascade risk.
- Forest-management effects propagate seaward, not into the eastern dune slacks.
- Both interventions are second-order against coastal-retreat drawdown.

Keywords: dune slack; groundwater; clearfell; BACI; restoration; coastal dune hydrology

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1. Introduction

Coastal dune slacks are among the most species-rich and most threatened groundwater-dependent habitats in north-west Europe. Their characteristic plant communities occupy a narrow band of water-table depth: the seasonal summer minimum must remain shallow enough to wet the rooting zone yet deep enough to avoid permanent inundation, a window of roughly 0.4 m that separates the wet-slack and dry-slack communities recognized by Curreli et al. (2013). A complementary indicator based on mean spring rather than summer water level has more recently been proposed (van Willegen et al., 2025); we frame the present analysis on the summer minimum, which is the more direct measure of the drought constraint on slack viability and the metric against which both interventions are evaluated here. Where the summer water table falls progressively, slack vegetation transitions toward dry-grassland and scrub, and the conservation interest is lost. At many British and continental sites this decline has prompted active intervention, most commonly the removal of conifer plantation established on stabilized back dunes during the twentieth century, and the mechanical lowering of the ground surface to bring the water table back into the rooting zone.

The hydrological rationale for these interventions is intuitive but rarely tested against a long record. Plantation removal is expected to recover the rainfall intercepted by the canopy and so to raise recharge; ground-surface lowering is expected to reduce the depth from the surface to the water table directly. Neither expectation, however, distinguishes a change in the mean water level from a change at the summer minimum, and neither accounts for the confounding trends — progressive climate drying and, at coastal sites, shoreline retreat — that operate over the same decades as the management. Without a control-based design and a long baseline, an apparent post-intervention improvement can reflect a wet run of years, and an apparent failure can mask a real benefit eroded by a deteriorating background.

Newborough Warren, on the south-west coast of Anglesey, offers an unusually strong setting for such a test. A continuous manual dipwell record spanning 2005–2026 covers 88 wells across a single sand body, two documented management interventions, and a coastline undergoing measurable retreat. The first intervention is a dune scraping carried out at one slack (CEH36) in April 2015; the second is an experimental clearfell of approximately 8 ha of Corsican pine (*Pinus nigra* var. *maritima*) in December 2017, undertaken on the recommendation of earlier researchers and implemented as an operational forestry action. The monitoring network includes wells inside, at the edge of, and well beyond the felled compartment, together with unfelled forest and open-dune controls, allowing the felling signal to be separated from the site-wide climate trajectory and from the coastal-retreat gradient.

This paper analyses both interventions within a Before–After–Control–Impact (BACI) framework and asks three questions. First, did either intervention raise the water table, and if so, in which season and by how much? Second, what mechanisms explain the response, and what do they imply for the durability of the gain? Third, how far do the effects of forest management reach across the site — in particular, can clearance in the forest zone benefit the eastern slacks where the ecological stakes are highest? The aquifer architecture, cluster structure and state-space model (SSM) on which the analysis rests are developed in the companion paper (Hollingham, 2026a); here we cite that framework rather than re-derive it, and concentrate on the interventions themselves.

2. Study site and interventions

2.1 Site

Newborough Warren is a calcareous coastal dune system of approximately 700 ha designated as a Special Area of Conservation for its dune slack habitats. A bedrock ridge forms the northern boundary of the study area; groundwater flows southward from this high ground across a single, highly transmissive body of clean dune sand. The drainage divides around a central dune topographic high: the forested western block discharges south-westward toward Caernarfon Bay, while the open eastern dune discharges south and south-east toward the Menai Strait.

This drainage geometry is central to the reach question addressed in Section 5.4: perturbations originating in the western forest propagate seaward, not into the eastern slacks. The geomorphology, the five-cluster partition of the monitoring network and the substrate contrasts are described in full in the companion paper (Hollingham, 2026a).

The monitoring record comprises end-of-month manual dipwell readings. For the analyses reported here the network is partitioned into five hydrogeological clusters — C1 Lake Edge (n = 7), C2 Dune (n = 24), C3 Western Residual (n = 21), C4 Main Forest (n = 9) and C5 Coastal Forest (n = 5), totalling 66 reference wells — with an extended network of additional wells used in robustness checks. The forest occupies clusters C4 and C5; the two carry the same Corsican pine canopy but differ in substrate, C4 sitting on thinner sand over irregular bedrock on the elevated ridge flank and C5 on deeper, more uniform coastal sand.

2.2 The scraping intervention

In April 2015 the ground surface of one dune slack, instrumented by dipwell CEH36, was mechanically lowered to raise the summer water table toward the surface. The excavation depth was not directly surveyed; back-calculation from the observed water-table response and the open-dune specific yield ($d\_ex = H_0 / Sy$) implies a lowering of approximately 0.42 m. A second scraping episode in October 2023 lowered two further slacks (CEH18, CEH21). CEH36 lies approximately 340 m seaward and downslope of the later clearfell footprint, a proximity that matters when the interaction between the two interventions is considered. The unscraped slack CEH4, and CEH22 nearer the coast, serve as paired controls.

2.3 The clearfell intervention

In December 2017 approximately 8.4 ha of Corsican pine was felled. The felled compartment straddles the gradational boundary between C4 Main Forest and C5 Coastal Forest, and the wells within and adjacent to it show mixed cluster affinities spanning C3, C4 and C5 (Hollingham, 2026a). The compartment cannot, therefore, be treated as a clean single-cluster experiment; its hydrological effects propagate downslope into the coastal catchment along the continuous drainage pathway. The sole in-situ Impact well is WMC3; four edge wells (CEH31, CEH20, CEH30, CEH16) sit closer to the felling boundary, and unfelled forest, coastal and climate controls complete the five-tier design described below. The dates that define the interventions — the December 2017 clearfell and the dune-scrape years — are the author's recollection, from twenty-one years monitoring the site; the one exception is the April 2015 scrape at CEH36, which is separately documented. No documentary record of the operations has been obtained from NRW or Forestry. These dates set the before/after split of the clearfell BACI.

3. Methods

3.1 BACI design and control tiers

Both interventions are evaluated as monthly BACI gaps — the water level at a target tier minus the level at a control tier — with the intervention "step" estimated as a level shift at the intervention date after adjustment for climate and spatial covariates. The clearfell analysis uses a five-tier classification: Impact (the felled well WMC3), Edge (felled-margin wells), Forest control (unfelled plantation), Coastal control and Climate control. Three counterfactuals are run for the clearfell — Forest, Climate and Combined — but the Forest control is the primary test, because it compares felled wells against unfelled forest sharing the same canopy and substrate and so isolates the canopy-removal response from the coastal-retreat gradient that contaminates comparisons against open-dune controls. The combined control dilutes the felling signal because it incorporates coastal-tier wells that carry the retreat signal, and the climate control spans a wider easting range still; both are reported for completeness but are not interpreted as clearfell effects.

3.2 ANCOVA specification

The clearfell step is estimated by ANCOVA fitted to the monthly BACI gap, with covariates: a centred cumulative water-balance (CWB) term capturing climate forcing; a distance-weighted scraping dummy (exponential decay length $L_s = 300$ m) that absorbs the 2015 scraping signal at wells near CEH36; a CWB × clearfell interaction testing whether climate sensitivity changed after felling; and a coastal-drift interaction that absorbs progressive divergence between treatment and control wells. The covariate is the fitted coastal-retreat profile evaluated at each tier's mean distance from the shore and differenced between target and control, scaled by that differential so its coefficient is dimensionless: a value of one is exactly the amplitude fitted at network scale. It is a drift control, not a measurement of a spatial gradient. Sensitivity of the clearfell coefficient to the scraping decay length was checked at $L_s = 200$, 300 and 500 m. The scraping intervention is analysed with a two-tier CUSUM-BACI framework: a Tier 1 correction removes background drift estimated at the controls, and a Tier 2 paired comparison estimates the residual scraping signal, cross-checked by synthetic control and by an SSM forward-residual method.

3.3 Coefficient decomposition and the shared SSM

To interpret the BACI steps mechanistically we draw on the state-space model developed in the companion paper, in which the monthly change in water level is partitioned into a rainfall-recharge term (β_1), an atmospheric-draw term (β_2, scaling PET) and a head-dependent drainage term (β_3, scaling displacement above a fixed drainage datum). For the intervention wells the SSM was fitted separately to the pre- and post-felling windows, yielding per-well coefficient shifts ($\Delta\beta_1$, $\Delta\beta_2$, $\Delta\beta_3$) and a predicted clearfell effect computed from those shifts. Canopy interception (24% of incident rainfall; Freeman, 2008) enters the forest clusters implicitly through their lower β_1 and is treated as a partition of the available PET energy budget rather than an additive loss (Hollingham, 2026a).

3.4 Scenario framework

Forward scenarios — full clearfell, 50% thinning and broadleaf conversion — were evaluated by perturbing the per-well β coefficients and computing the steady-state head response. Clearfell sets canopy interception to zero and adjusts β_2 at the forest wells using a BACI-corrected upper-bound multiplier; thinning halves the interception with a proportional adjustment; broadleaf conversion replaces the 24% pine interception with a 15% annual-mean deciduous value (Komatsu et al., 2011) and applies a seasonally varying β_2 multiplier that sits below the pine value through the leaf-off winter and above it during the growing season — Nov–Apr and May–Oct means of approximately 0.88 and 1.075 of the pine coefficient, an annual-mean effect close to unity — to reflect deciduous phenology. The framework aggregates per-well responses by inverse-distance weighting; it is a scenario aggregator, not a calibrated continuous-flow model, and applies no explicit propagation between clusters. All numerical results below are read from the live analysis pipeline.

4. Results

4.1 Dune scraping

4.1.1 Background drift

Both scraping controls show accelerating decline independent of any management action. The cumulative CUSUM at CEH4 reached −10.6 m and at CEH22 −19.0 m by early 2026, the convergent deterioration of two spatially separated wells confirming a progressive coastal-erosion signal at the western dune edge that any scraping benefit must be assessed against. The summer minimum at the CEH4 control deepened across successive management eras (era means −1.16 m at baseline, −1.18 m during 2015–17, −1.40 m post-felling), establishing the deteriorating background trajectory.

4.1.2 The CEH36 benefit

Three independent methods converge on the scraping benefit at CEH36 during the pure-scraping era (April 2015 – December 2017): a raw BACI step of +0.129 m against CEH4, a synthetic-control step of +0.137 m from eleven donor wells, and an SSM forward-residual step of +0.073 m (Figure 1). The convergence of the raw BACI and synthetic-control estimates with the near-zero SSM forward residual indicates a permanent ground-surface lowering — a step shift in the geometry — rather than a change in the aquifer's dynamic response. The benefit diminished but was not reversed during the post-felling era, as CEH4's own deterioration accounts for much of the apparent reduction in the BACI gap.

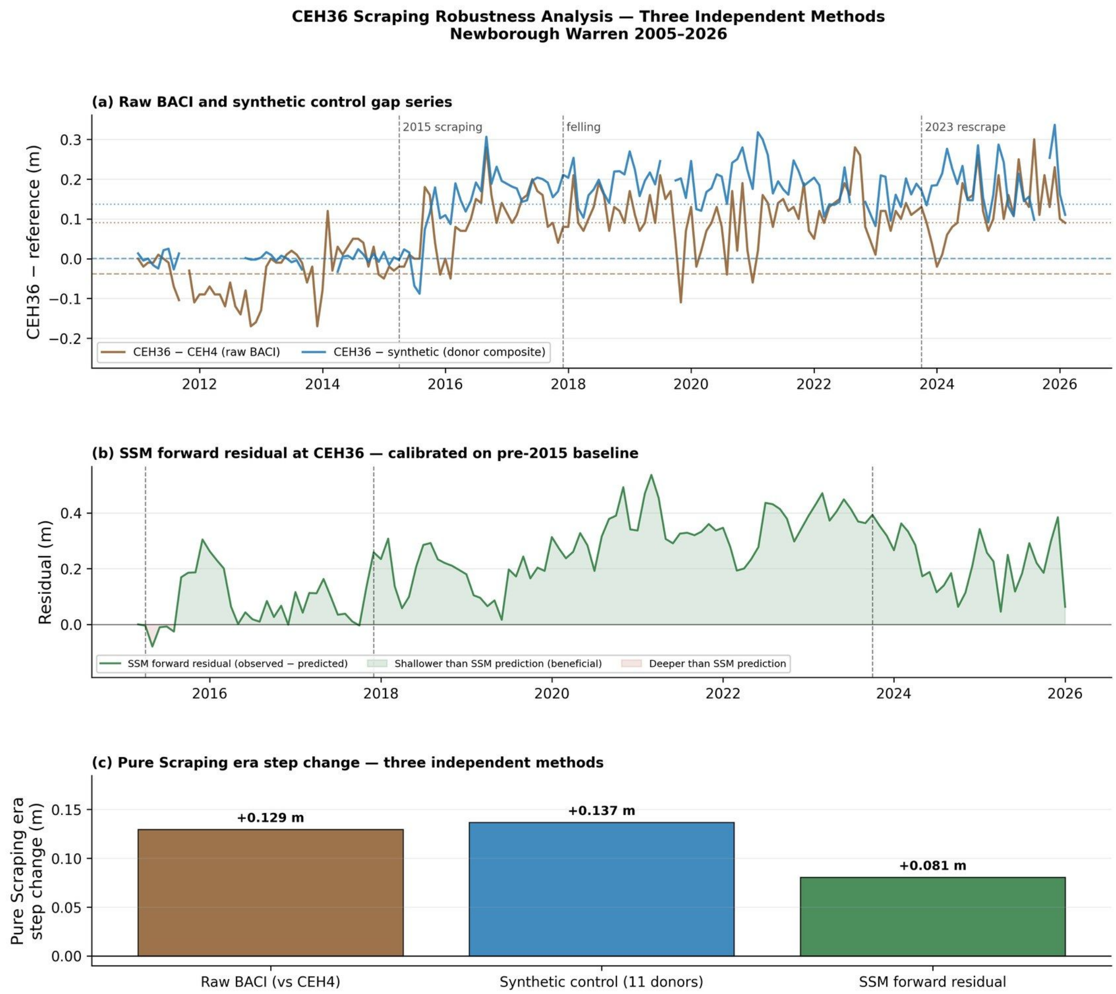


Figure 1. *Three-method robustness assessment of the scraping signal at CEH36 (measured). (a) Raw BACI gap (CEH36 minus CEH4) and synthetic-control gap, with intervention dates; the persistent positive offset after April 2015 marks the step change. (b) SSM forward residual, calibrated on the pre-scraping record; the persistently positive residual indicates a water table sitting above the pre-intervention model prediction. (c) Pure-scraping-era steps from the three methods: raw BACI +0.129 m, synthetic control +0.137 m, SSM residual +0.073 m. Source: 09_scrape_08_ceh36_robustness.*

The ecologically decisive result is at the summer minimum. The paired BACI summer-minimum analysis returns a significant improvement at CEH36 of +0.195 m against CEH4 ($p = 0.004$) and +0.161 m against the climate-control centroid ($p = 0.006$); the CEH4 control's own summer minimum deepened across successive eras over the same period (Section 4.1.1), confirming the deteriorating background against which the paired gain is measured. Neither of the slacks scraped in October 2023 shows a benefit that survives background correction (CEH18 +0.056 m, $p = 0.20$; CEH21 −0.008 m, $p = 0.89$), consistent with their short two-summer post-scraping record and, at CEH21, an unreliable control on its own coastal-erosion trajectory.

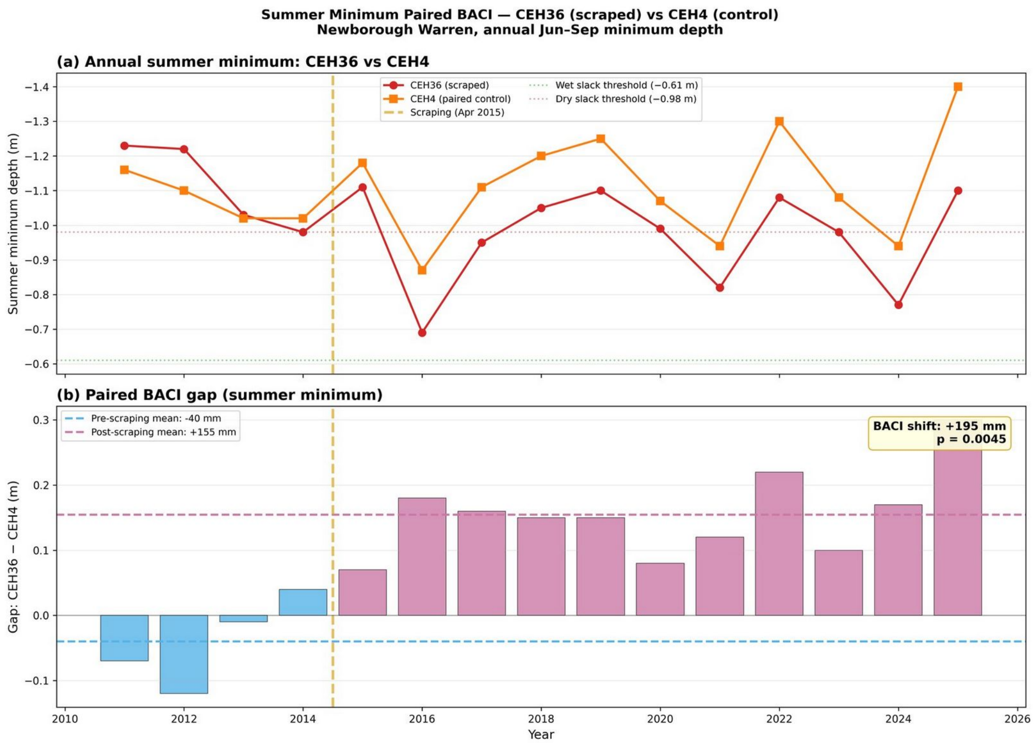


Figure 2. *Paired BACI summer-minimum analysis, CEH36 (scraped) versus CEH4 (control) (measured). (a) Annual summer minimum depth with the Curreli et al. (2013) ecological thresholds. (b) BACI gap (CEH36 minus CEH4) by year, pre- and post-scraping means; the paired shift is +0.195 m (p = 0.004). Source: 09c_04_summer_minima_paired.*

4.1.3 Drainage coefficient and propagation

Era-specific drainage coefficients (β_3) increased directionally at both treatment and control wells (CEH36 from 0.084 at baseline to 0.139 during pure scraping; CEH4 from 0.070 to 0.162 over the same period). Because the increase extends to unmanipulated wells, it is more consistent with SSM fitting artefacts at wells undergoing long-run erosion than with a genuine drainage response to scraping, and the per-era β_3 values should not be read as evidence of management-driven drainage change. A supplementary test for uphill propagation of the scraping signal was sensitive to the fitting method and is inconsistent with the aquifer diffusivity timescale, which at these distances exceeds the 31-month observation window; the propagation signal is best interpreted as a fitting artefact rather than a physical drainage response (Section 5.1.3). One near-field exception stands apart from this network-scale null. The displacement gap at WMC3 — the felled Impact well, some 260 m inland of CEH36 — fell by approximately 55 mm at the 2015 scraping and 54 mm at the 2023 re-scraping, near-identical across two independent events and the signature of a genuine relief-drainage capture at that one location; it is the same signal absorbed by the distance-weighted scraping covariate in the clearfell ANCOVA (Section 3.2). The distinction that matters is between this single, reproducible, measured near-field drawdown and a coherent network-wide distance-decay cone, which remains undetected: no smooth drawdown halo should be inferred beyond WMC3.

4.1.4 Scenario context at the scraped well

To place the measured benefit in context, alternative interventions were computed at CEH36 using its own SSM coefficients. The observed scraping benefit (+45.2 mm water-equivalent per month) substantially exceeds any modelled alternative at the same well: hypothetical clearfell +13.6, 50% thinning +6.8, broadleaf +2.9, and a wet-climate scenario +7.8 mm/month, against −14.4 mm/month under a dry-climate scenario. These modelled alternatives are equilibrium volumetric responses and cannot resolve a summer minimum, so the ecologically decisive comparison rests on the directly observed record: scraping delivered a measured paired-BACI summer-minimum benefit at CEH36 of +0.195 m against CEH4 (Section 4.1.2), a gain none of the modelled forest-management or climate alternatives approaches. This confirms that scraping operates through a fundamentally different mechanism, creating a topographic drain rather than modifying the surface energy balance.

4.2 The December 2017 clearfell

4.2.1 ANCOVA-BACI results

Against the Forest control — the primary counterfactual — the clearfell produced a significant rise in the mean monthly water table at the Impact well of +0.113 m (95% CI [+0.042, +0.183], $p = 0.002$, $R^2 = 0.24$, $n = 163$). The Edge-tier step was small and non-significant (+0.029 m, 95% CI [−0.021, +0.080], $p = 0.25$). The distance-weighted scraping covariate was significant in both Forest-control models (+0.324 m at Impact, $p = 0.004$; +0.568 m at Edge, $p < 0.001$), and clearfell coefficients were identical across the three scraping decay lengths tested, confirming that the scraping and clearfell terms are temporally well separated.

The Climate- and Combined-control comparisons are reported in Table 1. The Climate-control comparison returns a null (−0.014 m, $p = 0.56$), which should not be read as a weaker version of the Forest-control signal: against the open-dune

Climate control the felled area is statistically indistinguishable from open dune, so the forest hydrological signature has been removed and what remains is open-dune behaviour. This convergence is itself a positive finding — the Forest-control and Climate-control analyses together show what was gained (canopy removal raised the mean head +0.113 m relative to still-forested ground) and what was restored (in dynamical terms the felled ground's mean-level behaviour now matches open dune rather than forest). The Combined-control estimate (+0.069 m) lies between the two for the same geometric reason. All three comparisons are contaminated to varying degrees by the coastal-retreat gradient, which is why the Forest control is the primary test and Table 1 reports the others for completeness rather than as independent clearfell estimates.

The CWB × clearfell interaction was non-significant in all six models, indicating that canopy removal raised the mean water table without altering the system's sensitivity to climate forcing (Figure 3).

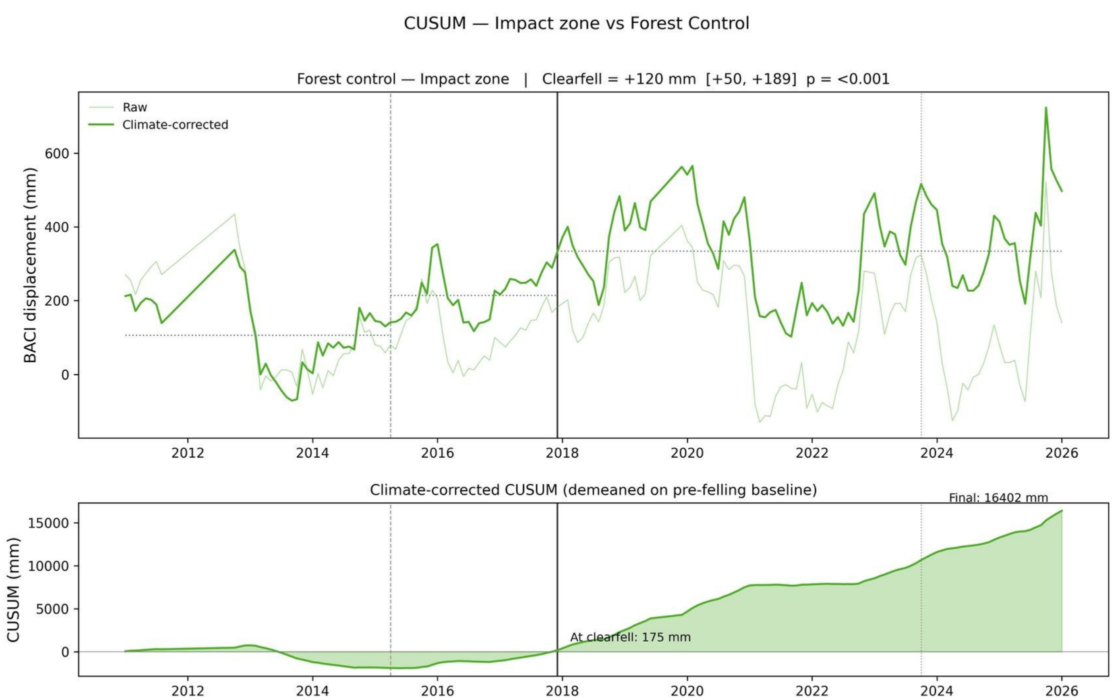


Figure 3. *Forest-control BACI time series and CUSUM for the Impact tier (WMC3) (measured). Upper panel: monthly BACI gap with climate-corrected series and era means. Lower panel: cumulative sum of the climate-corrected series demeaned on the pre-felling baseline; sustained positive accumulation after December 2017 confirms the clearfell step. Vertical lines: April 2015 scraping (dashed), December 2017 clearfell (solid), October 2023 re-scraping (dotted). Source: 10a_07_cusum_impact.*

The absorbed differential is not an independent estimate of the coastal gradient: the covariate is that gradient, so setting the two against each other is an internal consistency check rather than corroboration (Hollingham, 2026a). The coastal-retreat premise underlying the correction is independently corroborated by a model-free coast-to-inland transect, in which the coast-minus-inland head difference deepens at −28.2 mm yr^{-1} (95% CI [−34.2, −22.0]), within error of the network-scale gradient anomaly of −29.2 mm yr^{-1} recovered from an entirely independent construction (Hollingham, 2026a). The Edge comparison absorbs more drift than coastal retreat alone justifies, so any felling-attributable Edge signal is a lower bound rather than an inflated estimate.

4.2.2 Summer minima

No significant clearfell effect on summer minimum depth was detected at any tier. At the Impact well the shift was −0.007 m ($p = 0.91$); the pooled mixed-effects estimate at the Edge tier was −0.064 m and non-significant ($p = 0.12$). The distributional view by tier (Figure 4) shows the Impact and Edge tiers tracking the Forest-control trajectory in their summer extremes, with no detectable divergence after felling, while the Coastal control deepens progressively in a pattern consistent with coastal erosion rather than clearfell timing. The clearfell therefore registers in the mean monthly level but not in the summer extreme — the metric that determines slack viability — where felled and unfelled wells move together.

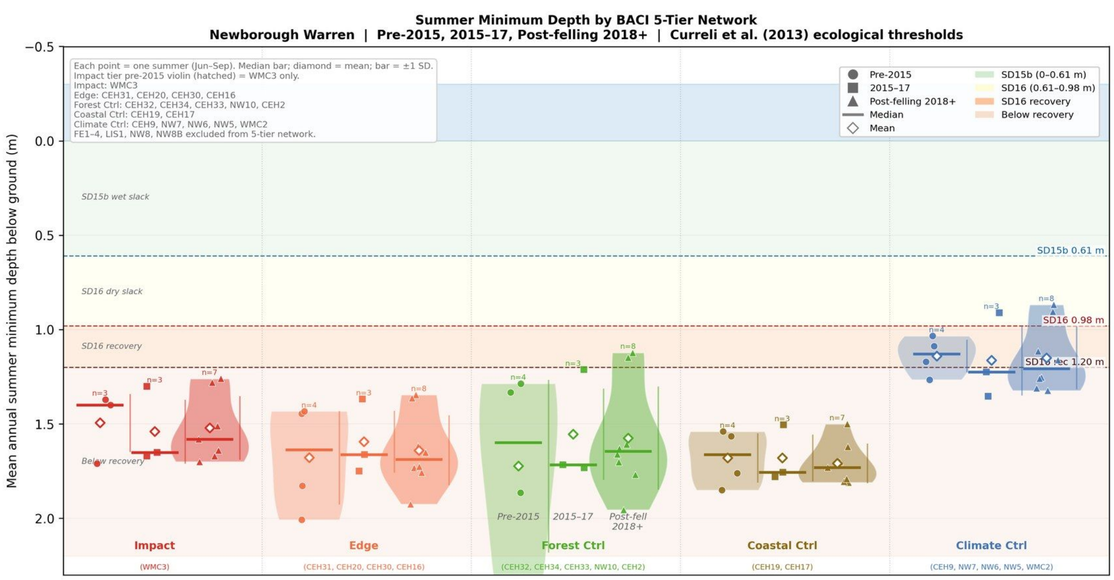


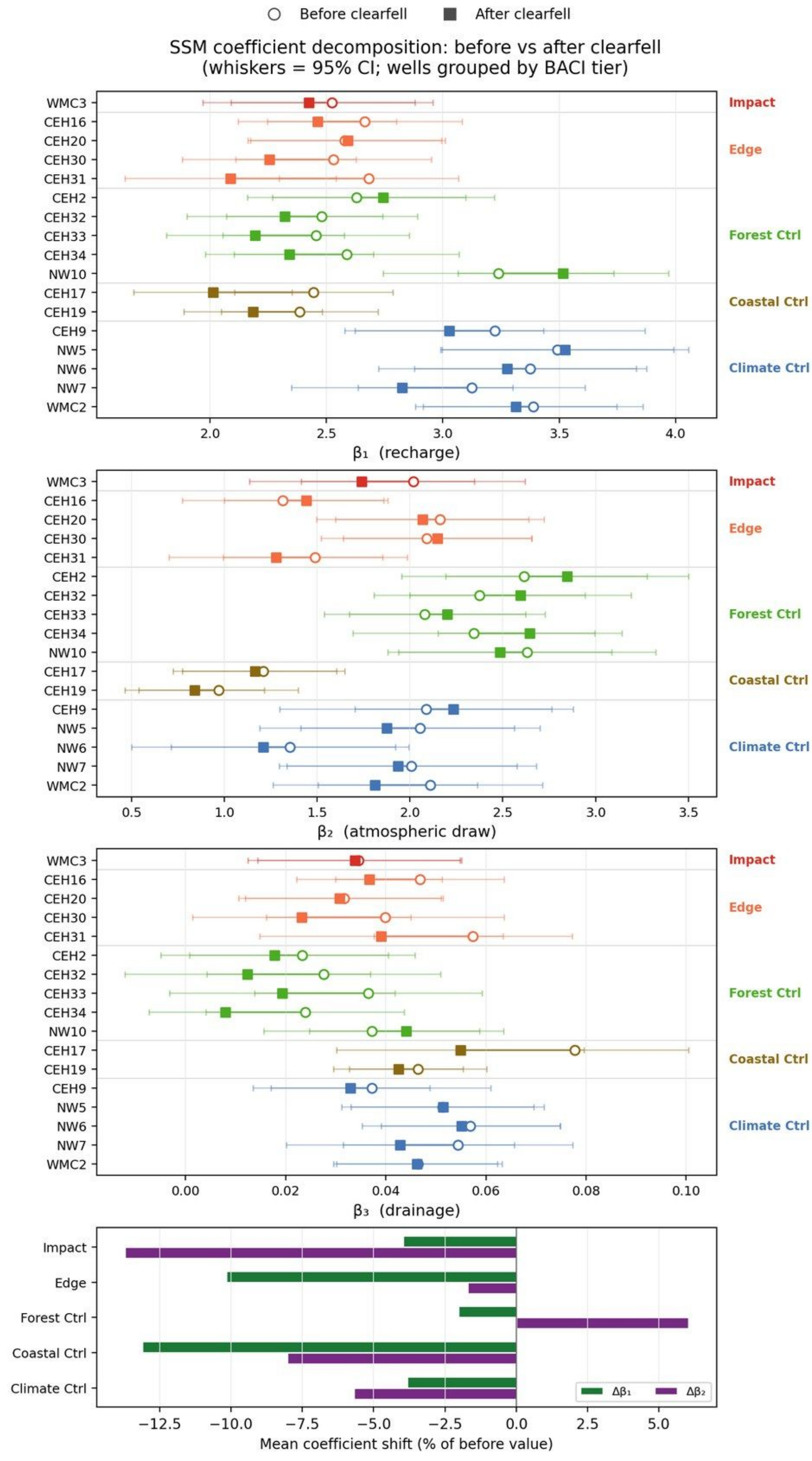
Figure 4. *Summer-minimum depth distributions by BACI tier across three management phases (measured). Violin width indicates frequency; diamond = mean, bar = median. The Impact and Edge tiers track the Forest-control trajectory with no detectable divergence in summer extremes after felling, despite the significant positive shift in the*

monthly ANCOVA; the Coastal control deepens progressively, consistent with coastal erosion. Ecological thresholds from Curreli et al. (2013). Source: 21_forestry_04_baci_zone_violin.

4.2.3 Coefficient decomposition

Fitting the SSM separately to the pre- and post-felling windows shows that recharge sensitivity (β_1) declined at nearly all wells across all tiers — a site-wide phenomenon, visible in controls and treatment wells alike, that rules out a clearfell-specific origin and is the strongest candidate mechanism for the observed summer-minimum decline (Figure 5). Atmospheric draw (β_2) moved in the opposite direction at the unfelled Corsican-pine Forest-control wells, rising by +0.06 to +0.27 (tier mean +0.169), whereas it fell at the felled Impact well (−0.31) and at the open-dune Climate and Coastal controls, and the broadleaf well NW10 fell rather than rose; within the Forest-control tier the β_1 and β_2 shifts are strongly anti-correlated ($r = -0.69$), though across the full network they are not ($r = -0.06$). Drainage (β_3) changes were small and spatially incoherent. Because β_1 falls comparably at controls and felled wells, a per-well reading of the shifts alone would imply no positive clearfell effect, in apparent tension with the significant positive ANCOVA step.

The two analyses are reconciled rather than contradictory. The ANCOVA measures the gap between felled and control wells, so the site-wide β_1 decline — which affects both — cancels and the canopy-removal effect becomes visible. The per-well decomposition measures coefficient changes at each well individually, where the β_1 decline registers as a negative contribution everywhere and, because controls and felled wells decline comparably, no clearfell-specific signal emerges. The decomposition therefore underestimates the felling benefit; it does not refute it.



Before clearfell
After clearfell
SSM coefficient decomposition: before vs after clearfell
(whiskers = 95% CI; wells grouped by BACI tier)
WMC3
CEH16
CEH20
CEH30
CEH31
CEH2
CEH32
CEH33
CEH34
NW10
CEH17
CEH19
CEH9
NW5
NW6
NW7
WMC2
Impact
Edge
Forest Ctrl
Coastal Ctrl
Climate Ctrl
β₁ (recharge)
β₂ (atmospheric draw)
β₃ (drainage)
Δβ₁
Δβ₂
Mean coefficient shift (% of before value)

Figure 5. *Before- and after-clearfell SSM coefficient estimates for the BACI-network wells, grouped and coloured by tier (measured). Open circles = before felling, filled squares = after, joined by a line; whiskers = 95% CI.* β_1 *recharge declines at nearly all wells across every tier — a site-wide signal.* β_2 *atmospheric draw rises only at the unfelled pine Forest controls while falling at the felled Impact well, the open-dune controls and the broadleaf well NW10 — an anomaly discussed in §5.3.* β_3 *drainage changes are small and spatially incoherent. Bottom panel: per-tier mean* β_1 *and* β_2 *shift as a percentage of the pre-felling value (*β_3 *omitted — its proportional swings are noise-dominated and are shown on its own panel above). Source: 10e_03_coefficient_shifts.*

4.2.4 Robustness

Four independent methods relax the ANCOVA assumptions. The six-donor synthetic control returns a clearfell step at the Impact well of +0.099 m ($p = 0.001$), estimated against a counterfactual built from donor wells outside the BACI network. The SSM forward-residual estimator, which calibrates each well on its own pre-scraping record and differences the post-felling against the post-scraping era, returns +0.056 m at WMC3 ($p = 0.001$). Both are positive and individually significant, but they are not on a common basis and neither is directly comparable to the ANCOVA's control-referenced step: the SSM residual at WMC3 sits below the mean step of the five Forest-control wells (+0.078 m, range +0.027 to +0.156 m), so on that estimator the felled well does not rise faster than the still-forested controls. The clearfell-specific component is therefore bracketed rather than converged upon by these two methods. Synthetic backward extension of the within-compartment wells (WMC3 + FE1 + FE2) returns a clearfell step of +0.080 m ($p < 0.001$). Decomposed, FE2 — sited inside the felled footprint — diverges +0.028 m post-felling ($p < 0.001$), whereas FE1, some 20 m outside the boundary in standing forest, shows no significant divergence (+0.009 m, $p = 0.14$): exactly the spatial pattern expected of a localized felling effect. A clearfell transect from plantation interior to felling core shows no significant distance gradient, though the post-felling spread in transect hydrographs widens, suggesting a distance-dependent response that seven wells lack the power to resolve.

4.3 Forest-management scenarios

The scenario framework quantifies the seasonal trade-off at the two forest clusters; it returns no response at C1, C2 or C3 under any forest-management scenario. Full clearfell produces a small positive net annual volumetric response at both forest clusters (C4 +8.3, C5 +10.5 mm/month), driven by the winter recharge gain from interception removal marginally exceeding the summer atmospheric-draw cost. Fifty per cent thinning gives roughly half this response (C4 +4.2, C5 +5.3

mm/month). Broadleaf conversion is near-neutral and differently signed across the two clusters (C4 −0.7, C5 +1.6 mm/month): the leafless-winter recharge advantage is offset by the growing-season transpiration penalty, slightly negative on the thin high-β_2 C4 substrate and marginally positive on the deeper C5 sand (Figure 6).

C5 captures a larger per-area clearfell response than C4 — about a quarter more (+10.5 against +8.3 mm/month) — despite carrying the same canopy, because its higher specific yield and lower β_2 attenuate the summer cost more strongly. This is the basis of an important caveat: the same canopy treatment delivers different per-area benefit across the forest, and the model predicts the *larger* response at C5 — yet the observed C5 trajectory is the steepest decline in the network. The model-versus-observation discrepancy at C5 is discussed in Section 5.2.3. All three scenario magnitudes are small relative to the climate-driven response over the same horizon, where the dry-climate scenario imposes −10 to −18 mm/month across the clusters.

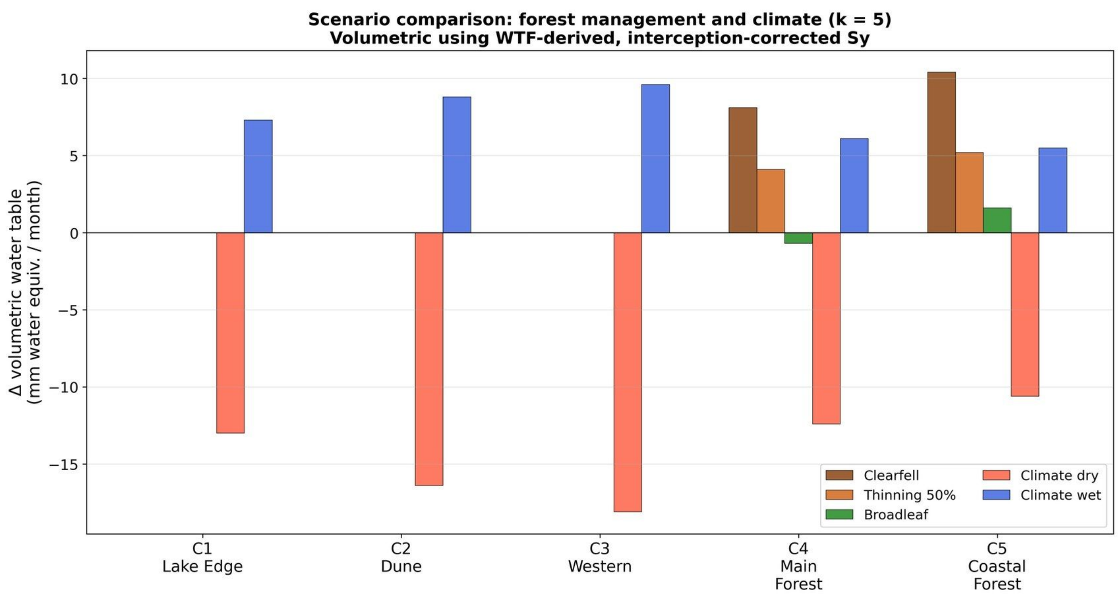


Figure 6. *Forest-management and climate scenario responses by cluster, expressed as volumetric water-equivalent change (mm/month) (modelled). Forest-management scenarios produce a response only at the forested clusters C4 and C5; clearfell and thinning are small and positive, broadleaf conversion near-neutral and differently signed across C4 and C5. Climate scenarios are spatially coherent and an order of magnitude larger. Steady-state predictions from the per-well SSM coefficient structure. Source: 21_forestry_05_scenario_comparison.*

Table 1. Three-counterfactual ANCOVA-BACI clearfell-step estimates (m) against each control tier, with the scraping covariate. The Forest control is the primary

test. Climate- and Combined-control rows are contaminated by the coastal-retreat gradient; the Climate-control null is discussed in the text as a convergence result rather than as an independent clearfell estimate. Values read from the live pipeline (`10a_report_numbers.csv`).

Control	Zone	Clearfell step (m)	95% CI	p	Scraping step (m)	Scraping p	R^2	n
Forest	Impact	+0.113	[+0.042, +0.184]	0.002	+0.324	0.004	0.241	163
Forest	Edge	+0.030	[−0.021, +0.080]	0.249	+0.569	<0.001	0.457	159
Climate	Impact	−0.015	[−0.063, +0.034]	0.555	+0.028	0.814	0.206	147
Climate	Edge	−0.107	[−0.167, −0.046]	<0.001	+0.289	0.456	0.241	144
Combined	Impact	+0.070	[+0.034, +0.105]	<0.001	+0.077	0.305	0.256	145
Combined	Edge	−0.020	[−0.046, +0.006]	0.129	+0.311	0.009	0.484	142

5. Discussion

5.1 Dune scraping

5.1.1 Measured benefit and durability

Scraping creates a topographic low that acts as a local drain, raising the water table at the scraped site by collecting lateral inflow. At CEH36 the benefit is unambiguous at the summer minimum — the +0.195 m paired improvement represents a shift across more than half the gradient separating the SD15b wet-slack and SD16 dry-slack community thresholds of Curreli et al. (2013) — and it persisted through the post-felling period even as the control deteriorated. Because the benefit is a geometric step rather than a change in dynamic response, the background climate-drying trend continues to operate at the scraped site as elsewhere, and the gain is slowly eroding. The two slacks scraped in 2023 have too short a record, and in one case too unreliable a control, to support any estimate; a re-scraping interval cannot yet be inferred from the record.

5.1.2 Coastal-erosion analogy and the cascade risk

The mechanism by which scraping delivers its benefit — lowering the ground surface to reduce the depth to water — is physically identical to the mechanism of coastal erosion. Both reduce ground elevation relative to a fixed absolute water-table position, both raise the displacement head at adjacent locations, and both consequently steepen drainage gradients away from the lowered zone. The accelerating CUSUM decline at the two coastal controls shows this process operating naturally at the western shoreline. The SSM drainage term operates in both directions, so a scraped zone with elevated displacement head draws water from its surroundings: scraping near the coast steepens the gradient between coast and interior and can deepen the water table at wells uphill.

This carries a management implication the physics make explicit even though the 31-month observation window cannot test it directly. If coastal scraping deepens summer minima at interior wells, those wells may in turn be identified as scraping candidates — lowering their surface, raising their displacement head, and propagating the drainage enhancement further inland. Each successive scrape addresses the local symptom while intensifying the systemic cause. This “chase-inland” dynamic is the topographic analogue of the apparent drainage acceleration that the clearfell analysis identifies as a fitting artefact: the aquifer is not gaining storage-decay capacity, but the systematic lowering of ground surfaces produces a head trajectory that the manager may read as progressive deepening requiring further intervention. On the precautionary principle, the spatial planning of any scraping programme should avoid amplifying the existing coastal drainage gradient; the most productive targets are the eastern transitional slacks (Section 5.4), not the coastal margin.

Placing the interventions on the same site-integrated footing as the background drivers characterized in the companion paper makes the cascade concrete rather than hypothetical. A scrape acts as a relief drain: it collects lateral inflow from the surrounding aquifer, and that redistributed inflow is precisely what raises the on-site slack level. The redistribution is real — it is the mechanism of the benefit — but it does not net to zero, because the surrounding drawdown footprint depletes more water than the slack recovers. The near-field component of that footprint is measured and reproducible (the −55 mm at WMC3 across both scrapes, Section 4.1.3); its wider extent is modelled rather than observed, and no smooth cone should be read into it beyond that one evidenced point. On the resulting first-order footing the net effect of scraping, integrated across the site, is a water-table loss of order 28% of the coastal-retreat drawdown in magnitude (Hollingham, 2026a) — a real secondary contributor to site-wide decline, not a local redistribution that balances. This does not diminish the value of a well-sited scrape, whose local summer-minimum gain is unambiguous; it means the benefit is genuinely local and is purchased against a modest site-wide cost, which is exactly why siting matters and why the coastal margin — where the off-site drawdown would reinforce the existing erosion gradient — is the wrong target.

5.1.3 Timescale

The propagation timescale is governed by the aquifer's hydraulic diffusivity and, for the relevant distances, is measured in years to decades — longer than the 31-month scraping observation window. This explains why no distance-decay in the drainage signal was detectable uphill of CEH36, accounts for the slow equilibration of any clearfell recovery, and means that the coastal-erosion signal now visible at the controls is a lagged response to earlier retreat: the present CUSUM trajectories are a leading indicator of inland conditions rather than a real-time measure of them. Monitoring programmes must be designed around this lag.

5.2 The clearfell experiment

5.2.1 A winter-recharge intervention

The clearfell is most directly tested against the in-situ Forest control, where it produced a significant +0.113 m rise in the mean monthly water table. Four further estimators relax the ANCOVA's assumptions, but they do not form a converging set: each is built on its own counterfactual and its own reference era, so none is like-for-like with the control-referenced ANCOVA step. Two of them — the six-donor synthetic control (+0.099 m) and the SSM forward residual at the felled well (+0.056 m) — are positive and individually significant; the SSM residual nonetheless sits below the mean step of the five Forest-control wells (+0.078 m), so on that estimator the felled well does not rise faster than the stands left standing. The within-compartment backward extension is more informative about where the signal sits than about how large it is: the post-felling divergence is significant at the well inside the felled footprint and absent at the well some 20 m outside it in standing forest, which is the spatial pattern a localized felling effect would leave. The transect adds no distance gradient that seven wells could resolve.

One sensitivity bears on how the coastal confound enters the step, and its direction is worth stating explicitly. The felled compartment sits closer to the eroding shoreline than the forest controls — about 550 m against 950 m — and the network-scale coastal-retreat gradient predicts a differential of roughly 10 mm yr^{-1} between them. That differential is not left uncorrected: the coastal-drift term in the ANCOVA absorbs it and, fitted freely, does so at 1.59 ± 0.44 times the network amplitude — indistinguishable from feeling it in full ($p = 0.18$) and firmly distinguishable from feeling none of it ($p < 0.001$). The +0.113 m step is therefore measured net of the coastal gradient rather than bounded by it; constraining the term to exactly the network amplitude, which the record does not require, lowers the step to +0.077 m ($p = 0.002$), reported as a sensitivity rather than the headline. The network gradient itself is not applied as a per-well correction: it is fitted with the clearfell wells withheld, accounts for under a fifth of the variance in per-well trend, and individual wells depart from its profile by more than the differential it predicts.

Taken together the robustness set is consistent with a genuine clearfell-specific component whose magnitude is bracketed rather than pinned down. The seasonal partition is the firmer result: the summer minima show no significant change at any tier. The honest synthesis is that canopy removal raised the mean — chiefly the winter level — without relieving the summer drought that governs slack viability. We resist a "success" or "failure" framing: the clearfell did what canopy removal mechanically can do at this site, which is recover part of the winter interception loss, and did not do what was hoped, which is lift the summer floor.

The seasonal partition follows from the canopy acting simultaneously as a hydrological sink and as a microclimatic shield (Section 5.3). Removing it recovers the interception loss, raising β_1 and winter recharge, but also raises β_2, so that a given summer PET produces a larger drawdown. The two effects operate in different seasons; their near-cancellation at the summer minimum, combined with the site-wide β_1 decline that deepens summer lows at felled and unfelled wells alike, leaves the summer floor essentially unmoved while lifting the mean. The non-significant CWB × clearfell interaction is consistent with this: the felling shifted the baseline from which the water table responds to climate, rather than the rate of that response.

An empirical comparison of five-year mean spring water levels (MSL5; van Willegen et al. 2025) between the pre-clearfell baseline (window end 2017, springs 2013–2017) and the current period (window end 2023, springs 2019–2023) shows no spatially distinct clearfell signal above the ±25 mm measurement noise threshold at either the Impact well (WMC3, +0 mm normalised) or the Forest Control tier (−48 mm, consistent with the site-wide secular deepening trend). This is consistent with the finding that the clearfell recovery is expressed in the monthly-mean water level rather than in spring peaks; the primary monthly ANCOVA-BACI result (+0.113 m, $p = 0.002$) remains the robust estimate of the clearfell step. The window ending 2025 was examined but excluded owing to artefactual gains at C4 Forest Control wells driven by differential drainage memory of the anomalously wet 2024 spring; see Hollingham (2026a) §5.7.5 for the window-selection rationale and three-check validation.

5.2.2 Reinterpreting the historical baseline

Earlier attributions of water-table depression to the plantation (e.g. Betson et al., 2002) were made during the driest decade in the long RAF Valley rainfall record and without a climate-partitioned, control-based design. The present analysis shows that the canopy does suppress winter recharge — confirmed by the significant mean recovery after felling — but that its effect on summer extremes is small relative to the site-wide β_1 decline. The historical baseline is best read as a compound of forest influence and exceptional climatic dryness whose seasonal structure was not resolvable at the time.

5.2.3 The C5 model–observation discrepancy

The scenario framework predicts that clearfell raises the C5 water table more than C4's, because C5's lower β_2 means the interception gain outweighs the evaporative penalty there. Yet C5 shows the steepest observed decline in the network. We flag this discrepancy rather than explain it away. The reconciliation we find most plausible is mechanistic: the clearfell scenario is a *vertical* flux adjustment — a winter recharge gain set against a summer evaporative cost — whereas the observed C5 decline is dominated by a *lateral* geomorphological forcing. C5 sits on deep, freely draining sand directly against the retreating coastal boundary, and its high hydraulic diffusivity allows the lateral drawdown signal from coastal erosion to propagate inland efficiently, overriding the localized vertical interception gain that the scenario predicts. This does not rescue the prediction so much as locate the missing term: the per-cluster SSM represents vertical exchange but not the lateral coastal boundary condition, and the mixed felling zone straddling the C4/C5 transition compounds the attribution. The coastal-retreat signal is the leading candidate. The companion paper's network-scale gradient analysis now places it on a quantitative footing: progressive coastal-boundary retreat accounts for approximately 48% of C5's observed summer-minimum decline, leaving an unexplained residual of order -20 mm yr^{-1} attributable to the thin-saturated-thickness and canopy-maturation mechanisms operating in parallel; the common-mode background contributes a small term of the opposite sign that the fit does not separately identify (Hollingham, 2026a). The discrepancy is therefore partly reconciled: the scenario framework over-predicts at C5 because the per-cluster SSM omits the coastal-retreat term that dominates C5's seaward boundary, not because the canopy mechanism fails. The precise 48% is fragile to which C5 well anchors the forest-free gradient fit, but the direction is robust — C5 is unambiguously the most coastal-affected cluster — and continued C5 monitoring with targeted coastal-retreat measurement remains a stated research priority.

5.3 Hydrological role of the plantation

The forest functions as both a sink and a shield. As a sink it suppresses recharge through 24% canopy interception (Freeman, 2008) and draws the water table down through transpiration; as a shield it intercepts PET energy at wet canopy surfaces before that energy can act on the water table. The shield is most clearly read not from the post-felling coefficient shifts but from the cross-cluster β_2 contrast, which is dominated by substrate, not canopy. C4 carries the network's highest β_2 not because its canopy draws hardest but because its thin sand over irregular bedrock has the lowest effective specific yield, so a given evaporative flux produces the largest head change; C5, on deeper sand, has a much lower β_2 despite identical canopy (Hollingham, 2026a). The practical consequence is that canopy-attributable changes in β_2 must be measured against a position-matched control — precisely the role of the Forest-control tier here.

That control tier, however, behaved in a way the canopy framework does not readily explain: atmospheric draw rose at the unfelled pine controls even as it fell almost everywhere else. We are cautious about reading this as a real increase in forest evaporative demand. The rise is modest — about 6% of β_2 — and at every individual well the pre- and post-felling confidence intervals overlap, so it is not significant well by well. Three considerations argue against an ecophysiological reading. The stands are already mature (~75 years), so further canopy maturation is an implausible driver of a 6–12% step in transpiration sensitivity over a single decade. The shift is also specific to the pine controls — the leaf-off broadleaf well moved the other way and the open-dune controls fell — a pattern a uniform climate-driven rise in demand would not produce. And within the pine controls the β_2 rise is mirror-imaged by the β_1 decline ($r = -0.69$), the signature of an ordinary-least-squares fit trading variance between two seasonally anti-correlated regressors — winter rainfall against summer evaporative demand — at wells where the canopy buffers and lags recharge enough to make the two hard to separate.

A warmer-winter increase in canopy interception, physically reasonable for an evergreen stand, is consistent with the recharge (β_1) decline but cannot account for the β_2 rise. In the Thornthwaite scheme used here winter PET is small and essentially unchanged between the two windows — the entire PET increase falls in summer — and the scheme cannot represent the wind- and humidity-driven evaporation of intercepted water at all, so increased winter interception would register as suppressed recharge, not as stronger atmospheric draw. The remaining physical possibility is a summer storage effect — as the table declines into the thinner, lower-specific-yield C4 sand, a given evaporative flux produces a larger head change, inflating apparent β_2 — but this too is a property of substrate and a falling table, not of the canopy. We therefore treat the Forest-control β_2 rise as an artefact of fitting and substrate rather than a measured change in forest water use; consistent with the companion paper's finding that absolute β_2 values are upper bounds best read as rankings (Hollingham, 2026a), the decomposition is reliable for the direction of the β_1 decline but not for a quantitative reading of β_2.

A direct observational test of broadleaf conversion exists within the network, and it is best read as a phenological proof-of-concept rather than a spatial counterfactual. Dipwell NW10 sits in a block of mixed broadleaf established after a 1993 clearance. Its decoupled SSM coefficients capture the deciduous seasonal partition directly: a higher recharge sensitivity ($\beta_1 = 3.53$ against a C4 median of 2.34), consistent with a leaf-off canopy transmitting more winter rainfall, set against an atmospheric draw ($\beta_2 = 2.52$) comparable to adjacent pine, consistent with that winter advantage being offset by growing-season transpiration. Its climate-normalized summer minimum sits about +0.25 m shallower than the pine interior, and — the feature that most supports the scenario framework — this anomaly is diminishing as the canopy closes, converging on the near-neutral scenario prediction. The spatial reading is weak: NW10 is a single, positionally confounded well, and the spread between two pine wells in the same cluster exceeds the NW10 anomaly, so it supports an indicative reading at most — a maturing broadleaf canopy whose net annual hydrological effect approaches that of the pine it replaced, with a different seasonal partition. A robust species attribution would require several instrumented wells across the broadleaf block.

We also note a slower process that the water-table analysis cannot resolve but that bears on restoration: progressive *Dothistroma* needle blight is already thinning parts of the canopy, offering a natural experiment in canopy-density effects on β_2 that requires only sustained monitoring. It is flagged as a monitoring question rather than a finding.

5.4 The reach of forest management

A central management question is whether forest management in the C4/C5 zone could improve conditions in the eastern open-dune clusters (C1, C2, C3), where the ecological thresholds are most critical and the remaining wet-slack resource is concentrated. The evidence indicates it cannot. The scenario framework produces no detectable response at C1–C3 under any forest-management scenario; the BACI record shows no felling-specific response at wells outside the forest zone once common-mode climate variability is removed; and the post-felling transect is spatially uniform with no inland gradient. This isolation is not a framework artefact. The drainage geometry sends forest-zone perturbations south-westward toward Caernarfon Bay, not eastward into C1 and C2 (Figure 7), and the steady-state drawdown of the existing 75-year-old plantation is already negligible at the eastern wells despite the plantation having had its full lifetime to reach equilibrium. Any clearfell recovery would propagate outward on the same years-to-decades timescale and with a comparable, spatially limited footprint.

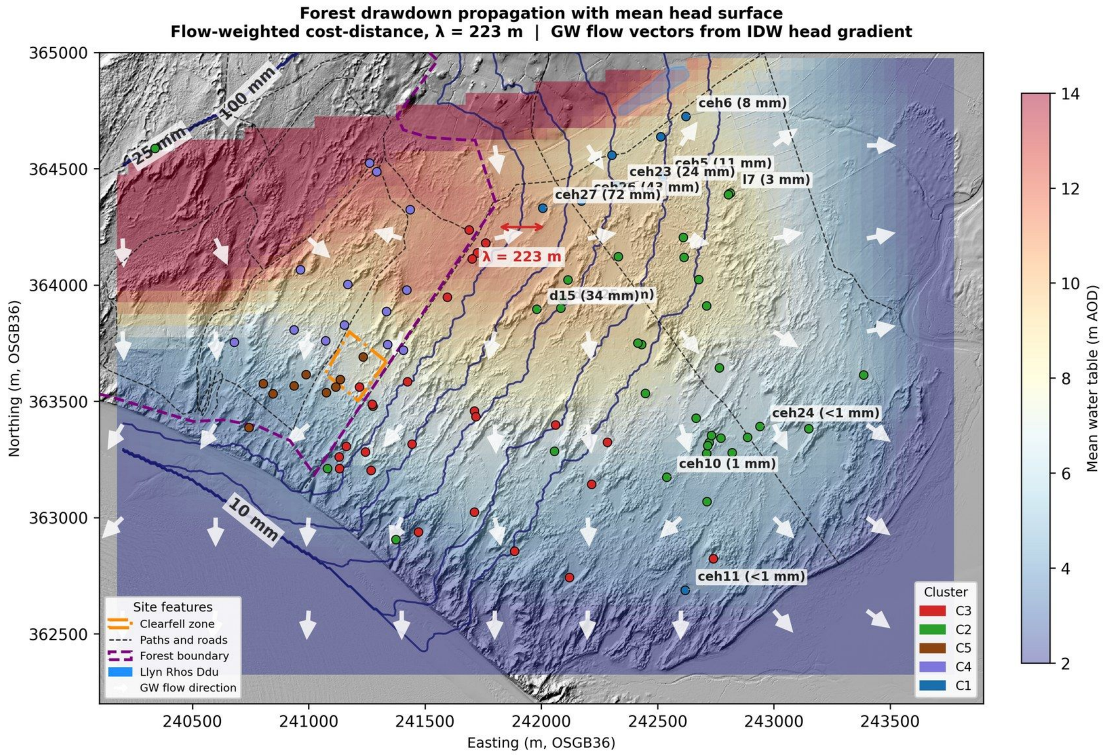


Figure 7. *Steady-state drawdown field of the existing plantation, illustrating the reach of forest-zone perturbations (illustrative — computed from an assumed canopy-edge head deficit, not a measured drawdown). The perturbation is directed south-westward toward Caernarfon Bay along the drainage gradient and is negligible at the eastern C1/C2 wells. The per-well inverse-distance framework is a geometric aggregator, not a calibrated flow model. Source: 20_drawdown_propagation.*

The clusters where the stakes are highest — C1 and C2, whose summer minima sit within 0.2–0.4 m of the community thresholds — therefore show no detectable response to forest management, and their summer trajectory is set by the site-wide climate signal. Because winter flooding across the western and forest clusters is controlled almost entirely by antecedent summer minimum depth rather than winter rainfall amount (Hollingham, 2026a), the primary restoration lever is summer minimum depth, over which forest management in the C4/C5 zone has no influence at C1 or C2.

5.5 Implications for restoration and monitoring

Three conclusions for management follow. First, summer minimum depth, not winter recharge, is the operative restoration target, which makes summer dipwell readings more diagnostically informative than winter peaks and reframes monitoring design accordingly. Second, topographic scraping is the most directly effective intervention in the record, but its benefit is local and its planning must respect the cascade risk by targeting the eastern C1/C2 transitional slacks — where a single scrape can bring a marginal well within reach of a mildly wet winter — rather than the coastal margin where it would amplify an existing erosion-driven gradient (Figure 8). Third, plantation clearance raises the mean winter water table but does not lift the summer floor and does not reach the eastern slacks; its hydrological case rests on winter-season and within-forest objectives, not on benefiting the wider slack resource.

Two points follow for monitoring design. First, the aquifer equilibrates to a step change over years rather than months — the lagged drainage adjustment that makes the clearfell's mean-level gain a transition toward a new state as much as a fixed outcome — so future BACI evaluations of topographic or canopy interventions should commit to multi-decadal baselines, lest a transient state observed in a short window be read as a permanent result. Second, because scraping steepens the inland drainage gradient in the same way coastal erosion does, any future scraping should be paired with a dedicated transect of dipwells running uphill along the drainage gradient from the worked slack, so that the chase-inland acceleration can be measured directly and pre-empted rather than reconstructed after the fact.

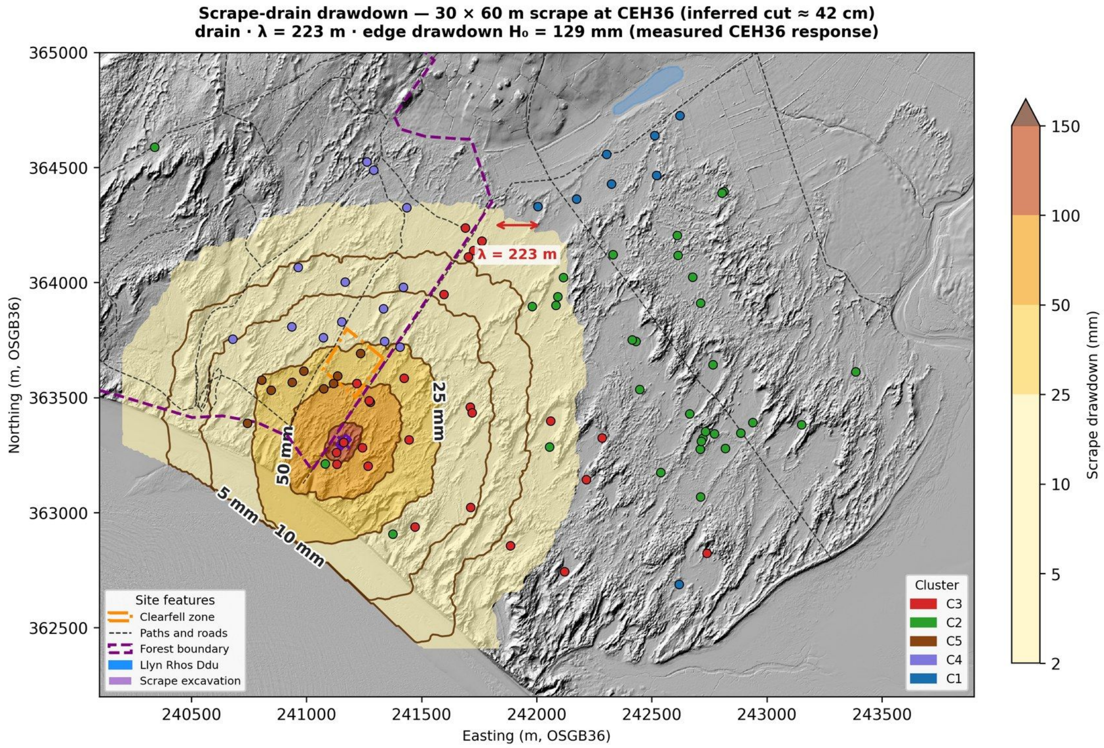


Figure 8. *Modelled steady-state drawdown field around the 2015 CEH36 scrape — edge magnitude empirically anchored, reach modelled. The edge drawdown is set to the measured CEH36 scraping response (≈0.13 m; Script 09a, Pure-Scraping era vs CEH4), not an assumed value; the surrounding field is a steady-state radial-drain decay along a flow-weighted cost-distance (leaky-aquifer length $\lambda = \sqrt{(K \cdot b/(Sy \cdot \beta_3))} \approx 230$ m), with the excavation interior masked because the slack itself rises. The excavation depth (≈0.42 m) is the inferred output, $d\_ex = H_0/Sy$, not a survey input. The elevated displacement head at the scrape steepens drainage gradients and draws the surrounding water table down — the same mechanism as coastal erosion, underlying the chase-inland cascade. Source: 20_scrape_drawdown_nohead.*

Underlying all three is the binding constraint: the summer drought trajectory, driven by a site-wide decline in recharge efficiency under a climate redistributing rainfall toward wetter winters and drier summers, is approaching the ecological viability thresholds within the next one to two decades. This ordering is borne out when the interventions are placed on a common site-integrated footing with the site's background drivers (Hollingham, 2026a). Coastal-retreat drawdown is by a wide margin the largest term on that footing; set against it, the net scraping loss is about 28% and the clearfell gain about 8%. The warren-wide common-mode component is not ranked alongside them, because the gradient fit does not identify it separately: its constant trades off against the trend in the cumulative-water-balance covariate, and on the committed fit it is statistically indistinguishable from zero (+0.18 mm yr^{-1}, standard error 0.54). The evidence that a site-wide climatic decline is the binding constraint rests on the recharge-efficiency (β_1) record rather than on that constant, and no equivalent-depth rate is quoted for it here. The two management interventions are, on this measure, second-order against a background they cannot arrest. The management challenge at Newborough is therefore hydrological adaptation — maintaining flooding capacity in an increasingly seasonal climate — rather than restoration to a prior state. The prediction equations, threshold maps and scenario framework underpinning these conclusions are available as interactive tools (Hollingham, 2026b).

5.6 Transferability to other dune systems

Newborough is a particular site, but neither intervention tested here is particular to it. Removal of conifer plantation established on stabilized back dunes, and mechanical lowering of the ground surface, are the two commonest active interventions in dune-slack restoration across north-west Europe, and the results reported here bear on both wherever they are proposed. Three findings generalize; a fourth constrains how far this site's numbers should be carried.

The seasonal partition of the clearfell response is the first. That canopy removal raises the mean water table without lifting the summer minimum is not a property of this compartment: it follows from the canopy acting simultaneously as a hydrological sink and as a microclimatic shield (Section 5.3), so that removal raises the recharge sensitivity β_1 and the atmospheric-draw term β_2 together. The two act in different seasons, and their near-cancellation at the summer minimum is what should be expected wherever an evergreen canopy is taken off a free-draining aquifer in a maritime climate. The practical consequence transfers directly. Where the restoration objective is expressed as summer minimum depth — as it is in the eco-hydrological thresholds of Curreli et al. (2013) and in most UK dune-slack designations — a clearfell should be appraised against a winter-season objective and not credited in advance with a summer benefit. Ainsdale on the Sefton Coast, where tree cover and a drying climate act on slack water tables in the same combination, is the closest British parallel (Clarke and Sanitwong Na Ayutthaya, 2010).

The second is scraping as relief drainage. The mechanism set out in Section 5.1.2 — a scrape raises the displacement head at its own position and therefore drains its surroundings, in the same manner as an eroding shoreline — is a property of the geometry, not of this site. It supplies a mechanism for a risk that management guidance already recognizes without explaining: that creating scrapes or pools can dry adjacent slacks (Buglife, 2019), a caution issued as a siting rule while scraping is otherwise treated as cut-to-the-water-table habitat creation (Grootjans et al., 2002; Jones et al., 2021). We are not aware of guidance that relates a scrape's geometry — its length, its orientation to the hydraulic gradient, and the base level to which it discharges — to its severity as a relief drain on the dune around it. Where scrapes are elongated and aligned with a coast-to-inland gradient, as they are here, that configuration is an efficient relief drain whatever its habitat intent, and the chase-inland cascade it can initiate is a design question anywhere the same geometry is used.

The third is the erosion analogue, which does not rest on this record alone. At Whiteford Burrows on the Gower coast of South Wales, the loss of some 4% of the spit width lowered the dune water table by approximately one metre (Robins et al., 2013). Direction and mechanism are what carry across, not rate: any dune system with an actively retreating margin should expect its interior water table to be falling for reasons management cannot influence, and should appraise its interventions on a footing that admits this.

What is most portable, finally, is the appraisal rather than any of the numbers. Reducing interventions and background drivers to a common site-integrated footing states a local gain against the site-wide cost that buys it (Hollingham, 2026a). On that footing coastal-retreat drawdown is the largest term at Newborough and both interventions are second-order; a sheltered or accreting system would rank differently, and it is the ranking exercise that transfers, not the ordering it produced here. The magnitudes do not transfer at all. The +0.113 m clearfell step is specific to one compartment, one control tier and one aquifer geometry, and is in any case measured net of the coastal confound rather than bounded by it (Section 5.2.1); the +0.195 m scraping benefit is a topographic step whose size is set by the excavation depth and the local drainage geometry. What another site can carry across is the seasonal partition, the drainage mechanism and the appraisal framework — not the coefficients.

6. Limitations

Neither intervention was randomly assigned, raising the usual possibility that pre-existing site characteristics confound the signal; the monitoring network was not designed as a purpose-built propagation transect and cannot resolve a residual felling signal within 100 m of the clearfell edge. Both intervention analyses treat the coastal-erosion confound through covariates and control selection rather than through explicit time-lagged modelling: the coastal-drift term captures the average spatial gradient but not the multi-year propagation of erosion signals from earlier retreat, so the contemporaneous management signal cannot be fully separated from the time-integrated coastal signal. The coastal-drift covariate is a drift control, and the comparison against the physics-based gradient model is an internal consistency check rather than independent corroboration, the covariate being that model's field (Hollingham, 2026a). What survives as independent support for the coastal driver is the model-free coast-to-inland transect. The coastal driver is thus established in direction and corroborated in rate, but its precise amplitude cannot be independently recovered from the monitoring network alone (Hollingham, 2026a), which bounds how sharply the contemporaneous management and time-integrated coastal signals can be separated. The Thornthwaite PET method overestimates evaporative demand in humid maritime climates, so the absolute scenario magnitudes are conservative upper bounds; the relative ranking of scenarios and the finding of structural isolation are independent of the PET method because they depend on coefficient ratios and spatial structure. The NW10 broadleaf comparison rests on a single, positionally confounded well with fewer than four years of post-canopy-closure record and should be read as indicative only. The spatial framework is a per-well scenario aggregator interpolated by inverse-distance weighting, not a calibrated continuous-flow model; resolving cluster-to-cluster coupling would require slug tests and a ground-penetrating-radar survey of aquifer thickness, identified as priority fieldwork. An empirical MSL5 change comparison (window end 2017 vs window end 2023) confirms the absence of a detectable clearfell or scraping signal in five-year spring means; the window choice requires careful validation at sites where the drainage recession time $1/\beta_3$ exceeds 12 months at some wells, as anomalous wet springs produce artefactual cluster-level gains in any window that includes them (Hollingham, 2026a §6.9).

7. Conclusions

A 21-year manual record and a five-tier BACI design allow the hydrological effects of two dune-restoration interventions to be separated from a deteriorating climatic and coastal background. The December 2017 clearfell raised the mean monthly water table at the felled well by +0.113 m against an in-situ forest control, but left summer minimum depth — the metric governing slack viability — unchanged at every tier; canopy removal is, at this site, a winter-recharge intervention whose effect on the summer floor is small against a site-wide decline in recharge efficiency. The 2015 dune scraping delivered a larger and directly ecological benefit, improving the summer minimum at the scraped slack by +0.195

m, but did so by permanently lowering the ground surface — mechanically identical to coastal erosion, and carrying a chase-inland cascade risk that, integrated across the site, registers as a small net water-table loss. Forest-management effects propagate seaward and do not reach the eastern slacks where the ecological need is greatest. The operative restoration lever is summer minimum depth, best addressed by carefully sited scraping in the eastern transitional zone, while the binding constraint remains the climate-driven summer drought trajectory — traceable, in the coefficient decomposition, to a site-wide decline in rainfall-recharge efficiency (β_1) as the climate redistributes rainfall toward wetter winters and drier summers. On a common site-integrated footing, coastal-retreat drawdown is the largest term and dwarfs both management interventions; the warren-wide common-mode component is not quantified by that analysis and is not ranked against them. The companion paper (Hollingham, 2026a) sets out the aquifer-characterization framework on which these conclusions rest.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used Anthropic's Claude (successive Claude Opus and Claude Fable models, accessed through the Claude desktop application) for substantial drafting and editing of the manuscript text: composing prose from the author's analytical outputs and decisions, structuring and restructuring sections, and correcting the text against the study's committed analytical outputs. The conception and design of the study, the twenty-one-year manual dipwell record on which it rests, the analytical decisions, and the interpretation of the results are the author's own. Numerical values in the manuscript were verified against the committed outputs of the analysis pipeline, and a decision log records the methodological and editorial calls made by the author during the work. After using this tool the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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